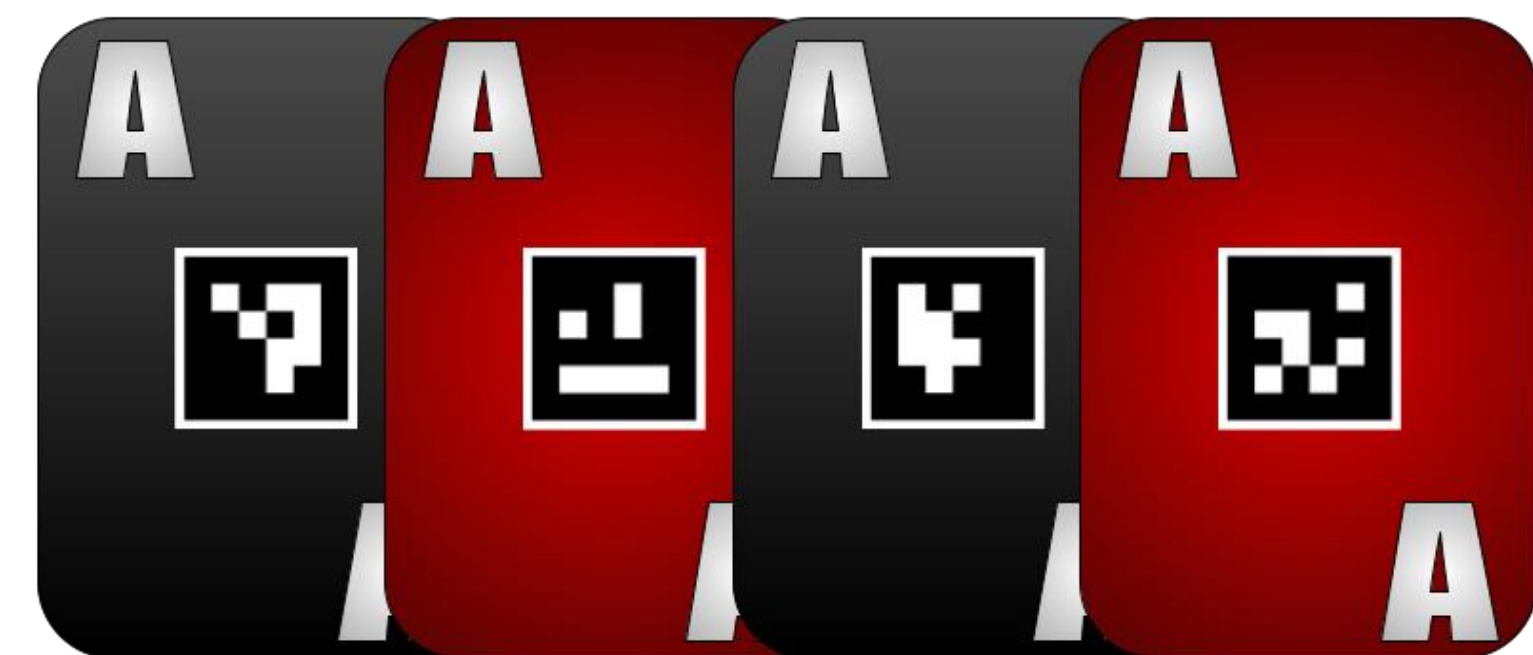




From Order to Chaos

Elijah Young

2.671 Measurement and Instrumentation

Abstract

Countless card games, from casinos to kitchen tables, rely on shuffling to ensure fair play, yet limited data exists on how different shuffle methods achieve true randomness. The effectiveness of riffle, overhand, and machine shuffling methods was quantified using statistical metrics derived from experimental trials by tracking how each card's position changed after every shuffle. A custom computer vision system using ArUco fiducial markers tracked all 52 card positions across 210 trials at shuffle counts $k = 0-7$. Randomness was evaluated through positional histograms, Adjacent-Pair Survival (APS), and normalized Total Variation (TV) distance. Riffle and machine shuffles approached the theoretical baseline of two surviving pairs with APS randomization rates of 1.49 and 1.56 per shuffle, respectively, while the overhand shuffle mixed more slowly at 0.55 per shuffle. Global TV analysis revealed decay rates of 0.53 for the riffle, 0.40 for the machine, and 0.23 for the overhand shuffle, indicating that full-deck randomness is achieved within roughly seven riffles but remains incomplete after seven overhands. These findings confirm theoretical predictions that riffle-style shuffles randomize far faster than overhand techniques, providing experimental evidence that links mathematical mixing theory to real-world shuffling behavior.

Shuffling Methodologies

Riffle Shuffle



[1]

Overhand Shuffle

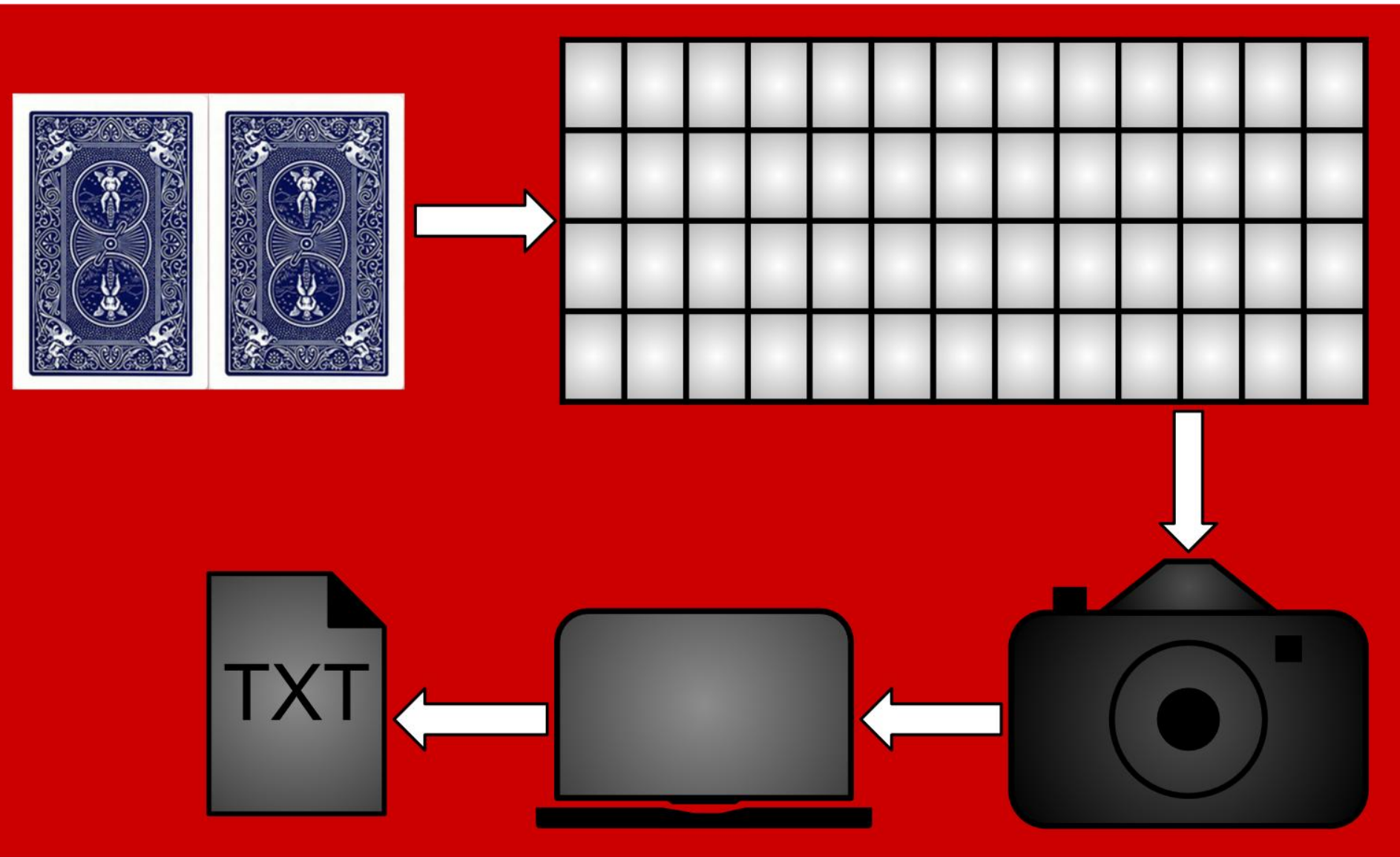


Machine Shuffle



[2]

Experimental Analysis



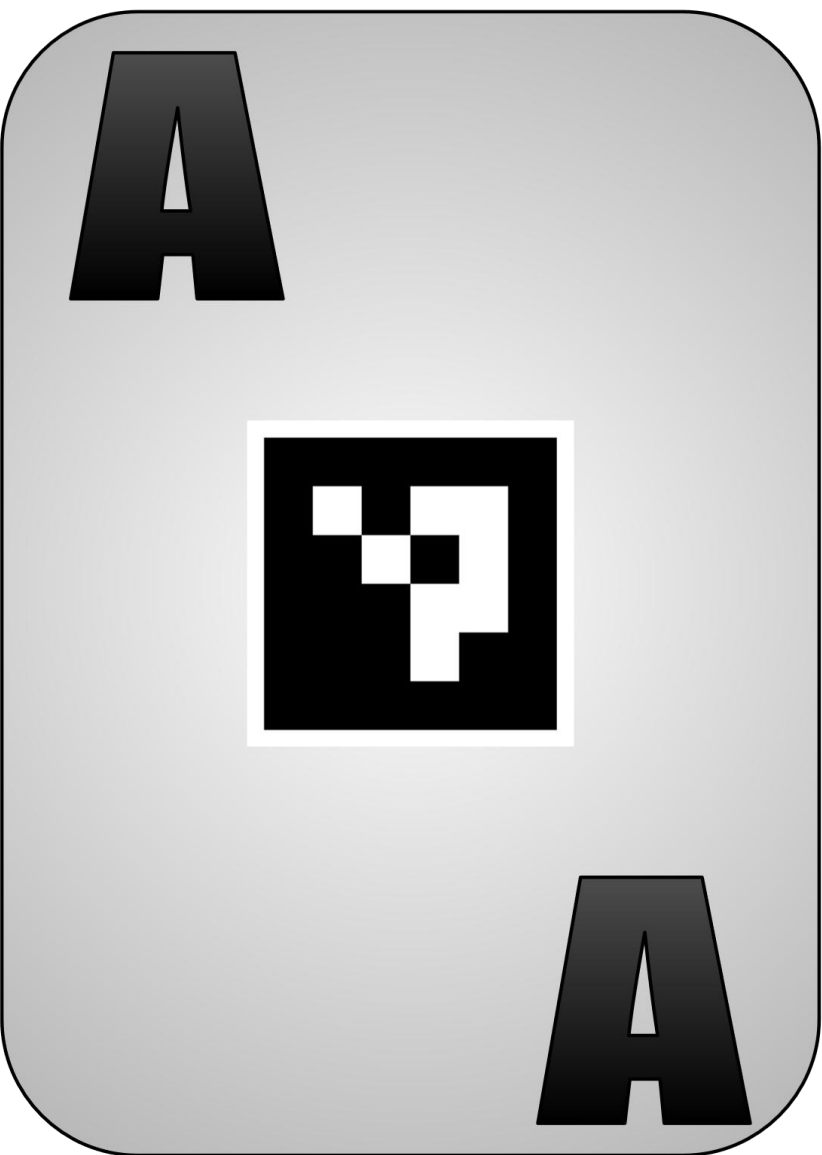
The cards are laid out in the post shuffle order in a 13 by 4 grid. The cards are then photographed. The python script scans the ArUco markers on the cards and using computer vision to create a txt file with the card order.

References

[1] Griffiths, S., 2015, "The Best Way to Shuffle Cards Revealed," Mail Online. [Online].

[2] INNCCN, "Product image of INNCCN X1," *Amazon.com*, 2024. [Photograph]. Available: <https://www.amazon.com/dp/B0F93FT6X1>.

ArUco Marker

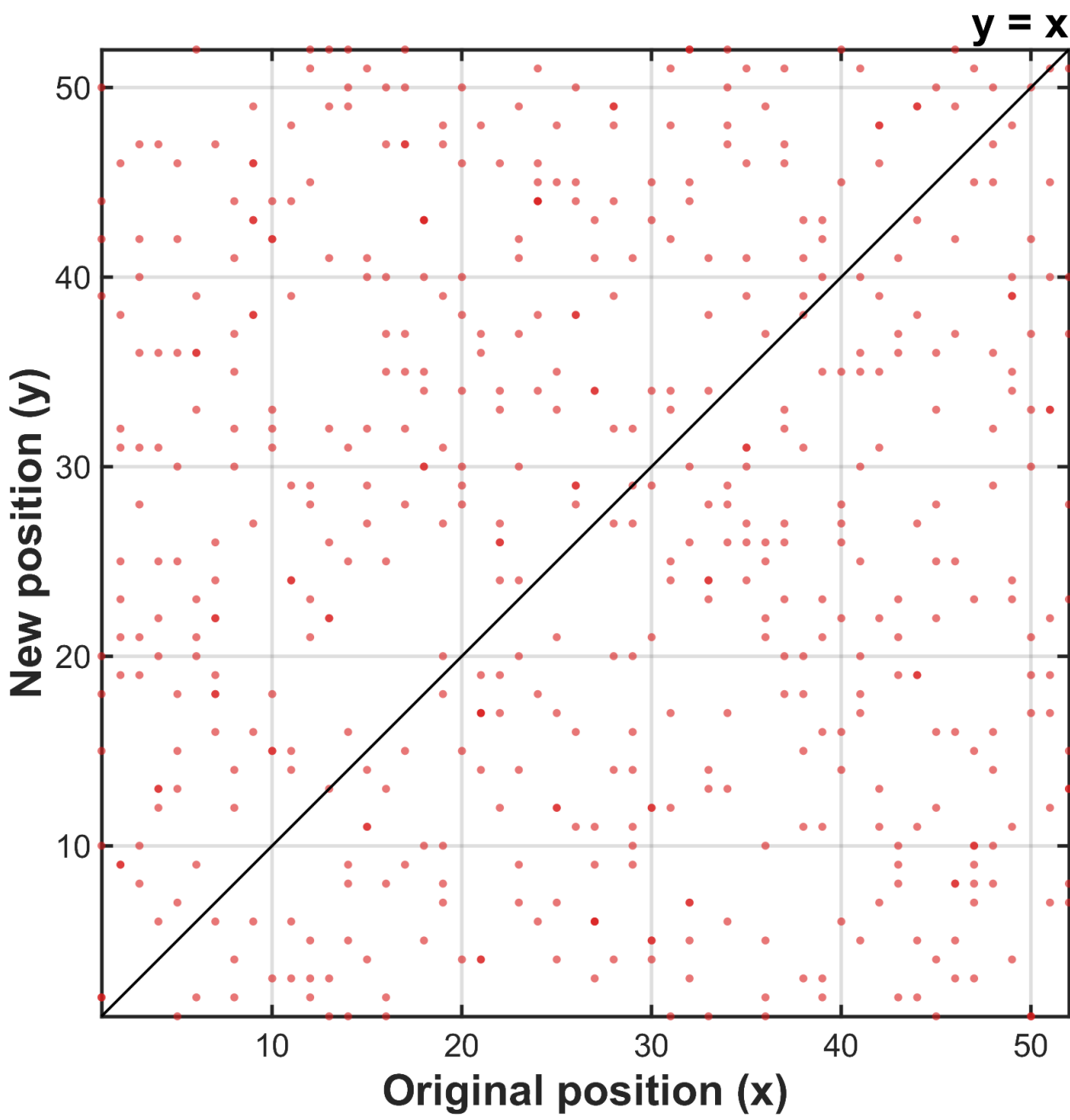


ArUco fiducial marker used to uniquely identify each card for automated computer-vision tracking.

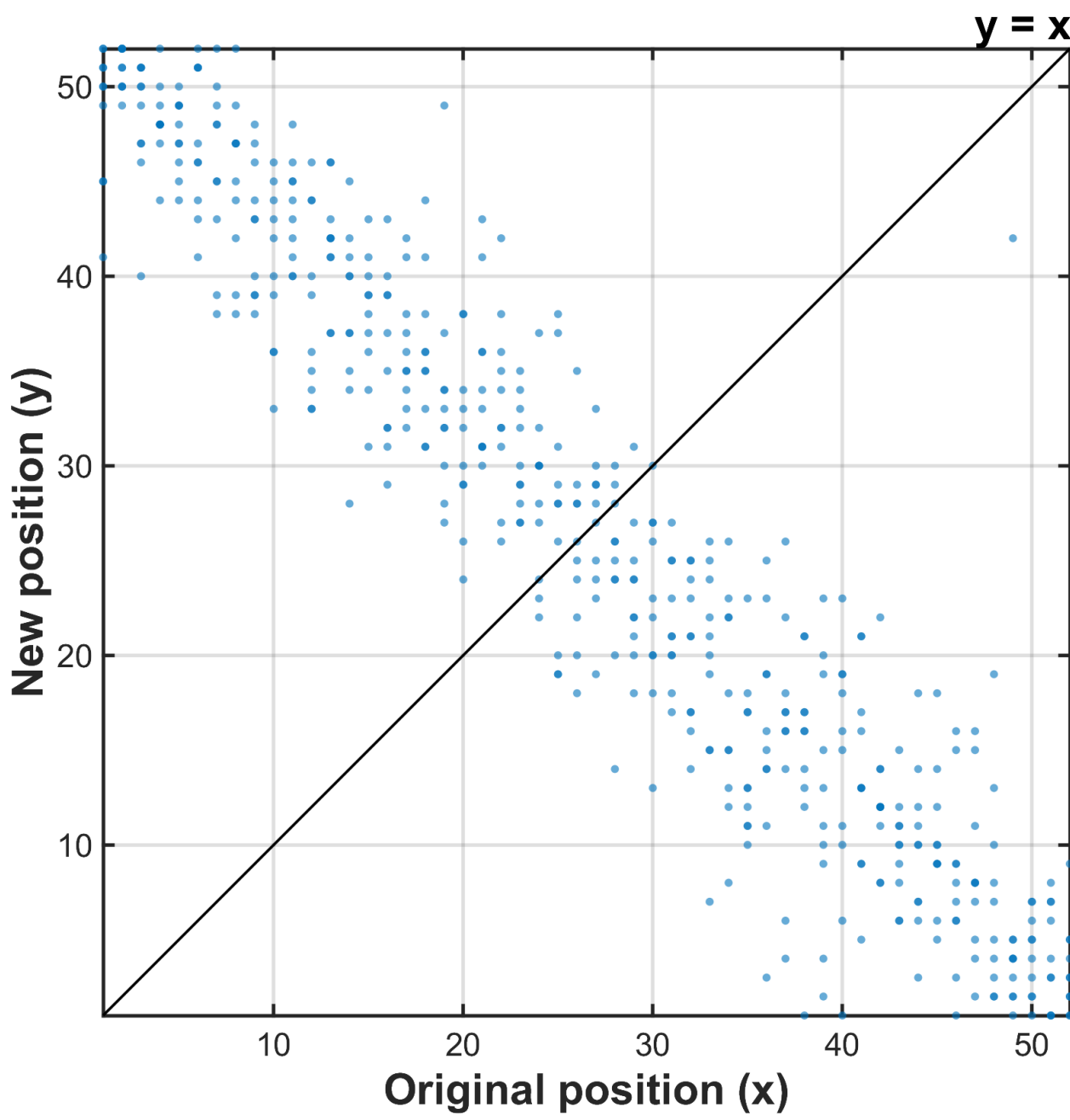
Acknowledgements

Special thanks to Danielle Knutson for her invaluable help in the development of the computer vision code and assistance with the photographic data collection. Another thanks to Dr. Huang, Dr. Bruner, and Dr. Hughey for their insights and support of the project.

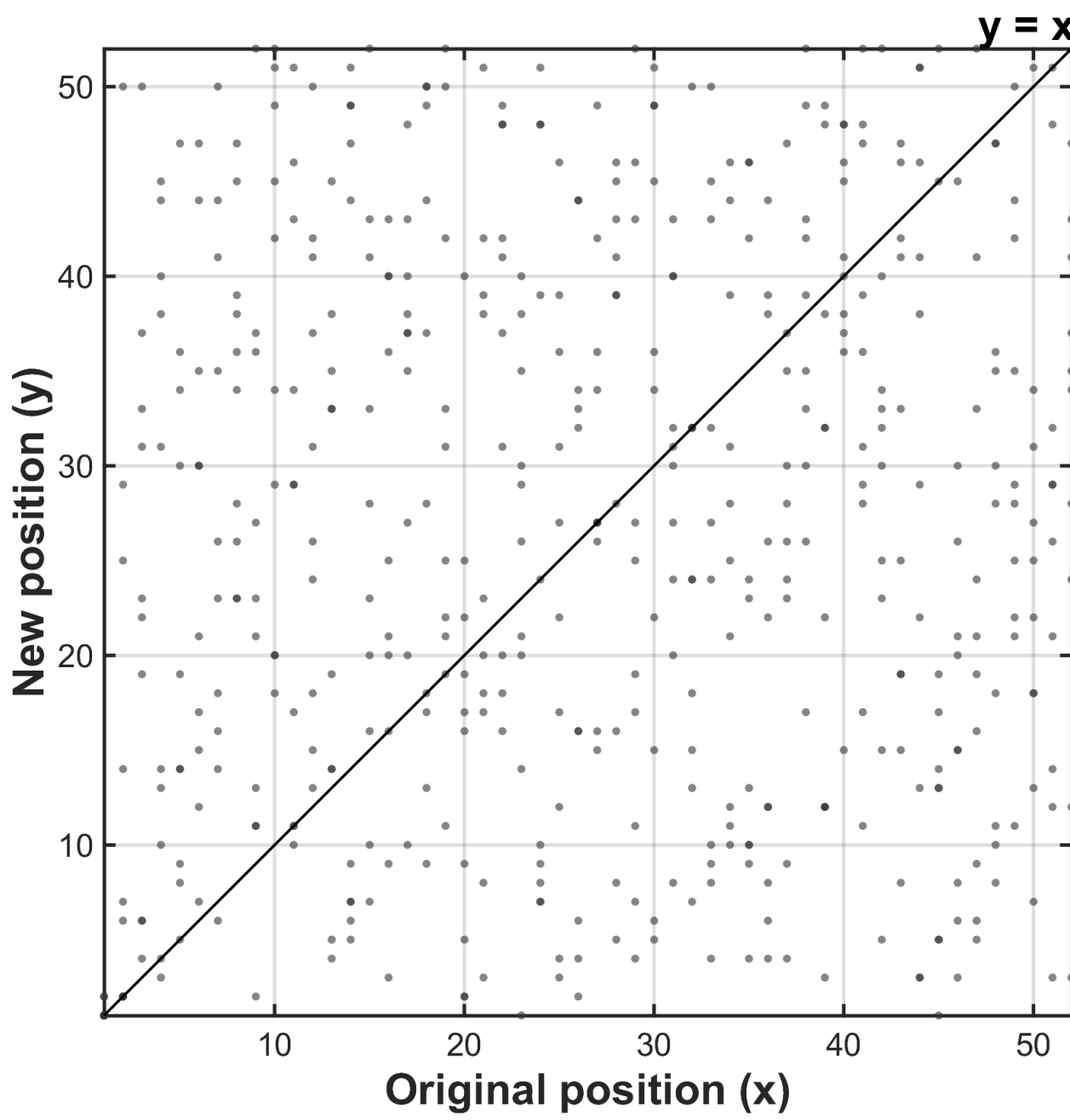
Positional Histograms



Riffle Shuffle k=7



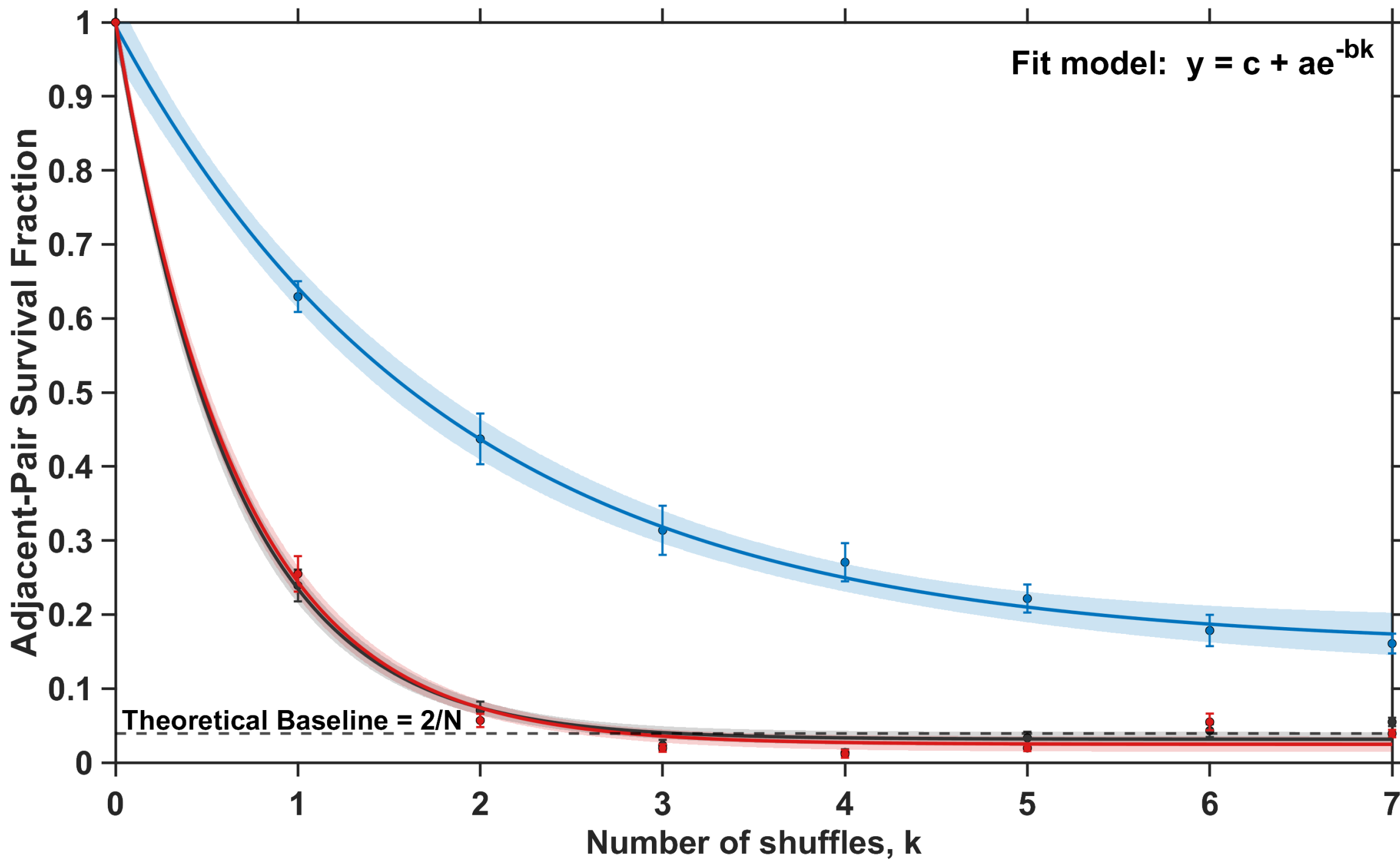
Overhand Shuffle k=7



Machine Shuffle k=7

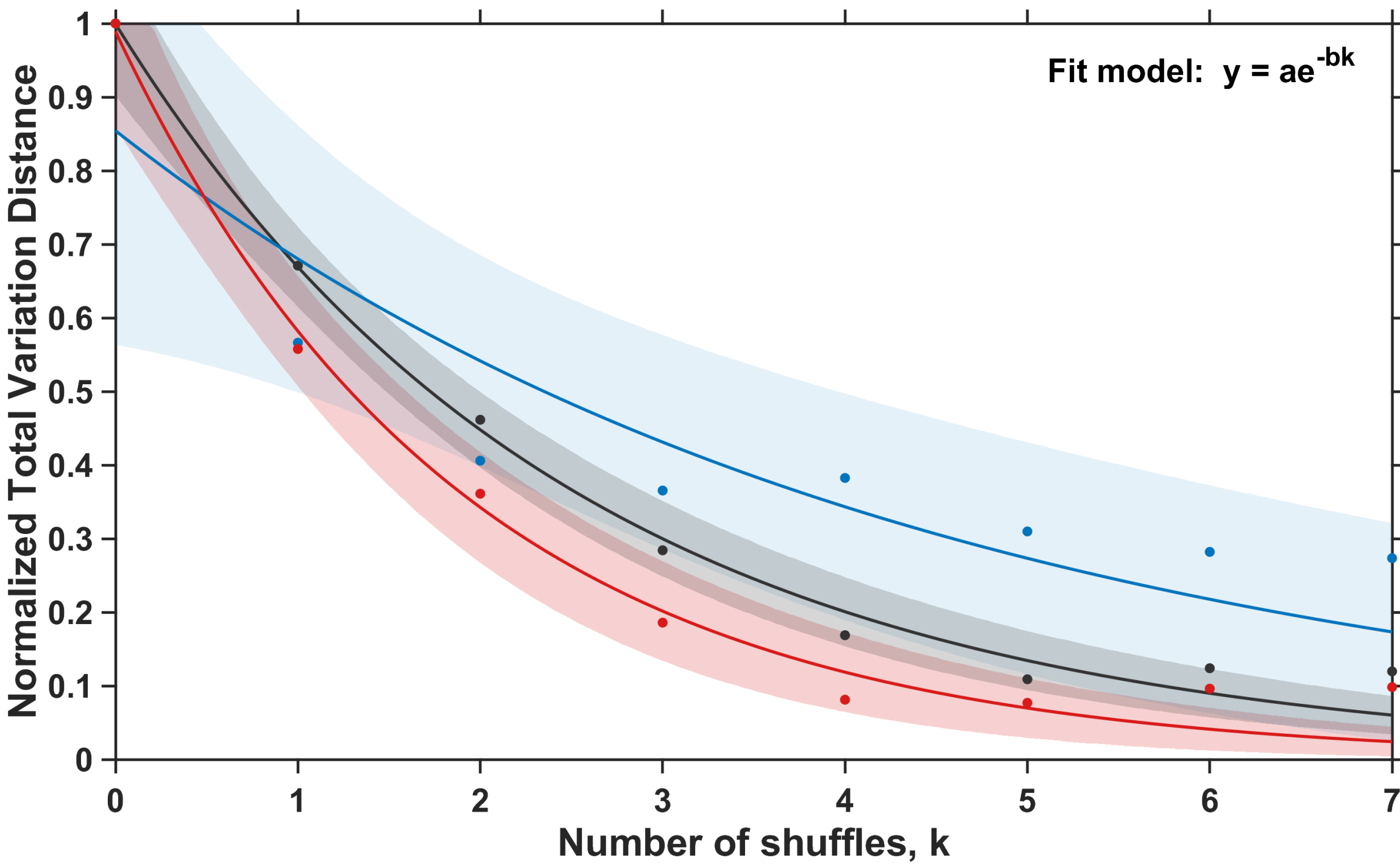
Adjacent Pair Survival

APS measures how many originally neighboring card pairs remain adjacent after shuffling.



Normalized Total Variation

Normalized TV measures how close the deck's card-position is to the uniform distribution of a perfectly random shuffle.



Conclusion

- Riffle and machine shuffles showed comparable randomization rates, with human riffles mixing slightly faster ($b_{APS} = 1.49$, $b_{TV} = 0.53$) than machine ($b_{APS} = 1.56$, $b_{TV} = 0.40$).
- Overhand shuffling was dramatically slower, with much lower decay rates ($b_{APS} = 0.55$, $b_{TV} = 0.23$) confirming its inefficiency for producing random decks.
- Computer vision with ArUco tracking proved highly effective, enabling fully automated measurement of both local and global randomness across 210 trials.
- Card aficionados should riffle shuffle at least 4 times and avoid overhand shuffles to ensure fair play and achieve sufficient randomization in their games.